

Personalized medicine

In medical arena, condition of each and every patient will differ. Hence, a system should be tailored to meet the requirements of each individual patient. This is where the importance of Personalized Medicine comes into picture.

The term Personalized Medicine was mainly derived from Pharmacogenomics (pharmacology + genomics), which examines the role of genetics in drug response. Patients often show genetics variations while a drug is being administered and it needs to be monitored with connected devices. This is what pharmacogenomics deals with by correlating gene expression or single-nucleotide polymorphisms with a wide range of parameters such as drug absorption, distribution, metabolism, elimination in addition to drug receptor target effects.



Genetic assessment through smart devices will help advancements in medicine

Personalized Medicine relies on technology to confirm patient's disease and diagnosis by testing various aspects such as fundamental biology, DNA, RNA or protein. With the help of technology, physicians make use of various connected devices such as blood pressure monitors, glucometers, wireless weight scales, ECG and EEG for patient diagnosis.

For example, if a patient is suffering from epilepsy, doctors should make use of wireless wearable EEG devices to monitor brain signals which doesn't require application of gels. These devices issue alerts via text or email as and when a patient shows changes which need medical attention. The appropriate treatment can then be administered based on the data captured from the devices.